

Smart Step Milestone #4 Requirements

This document describes the requirements for the **fourth development milestone** (Milestone #4) of the **Smart Step** application.

Milestone #4 focuses on introducing weekly activity reports and detailed analytics views. At this stage, the application is extended with a dedicated Weekly Activity Report screen that allows users to review their activity data by week, switch between different metrics (Steps, Calories, Time, Distance), and better understand trends and progress over time. This milestone builds on the tracking, metrics, and AI-driven insights introduced in the previous milestones.

The mockups define the overall appearance and expected behavior of the interface. Specific technical decisions (component structure, state handling, animation implementation) are left to the participants, as long as the final result follows the described logic and interaction patterns.

You can find the Smart Step **mockups here**:

<https://www.figma.com/design/aTiGCCavPLtgZNGXcV23H6/Smart-Step?node-id=67-2165>

Milestone #4 Goal

- Introduce the **Weekly Activity Report** screen with aggregated weekly data
- Display a **Top Summary Card** for the active metric and selected week
- Support switching between **Steps, Calories, Time, and Distance** via bottom metric tabs
- Define **active, inactive, and no-data states** for daily report items
- Provide adaptive layouts for **mobile and wider screens**

Icons

You may use Material Design icons where appropriate. If a suitable Material icon is not available, use the custom icons provided in the Figma mockups.

In Figma, any icon or image can be **exported** by selecting the element and clicking “Export” in the **right-hand panel**. In this panel, you can also choose the desired format (PNG, SVG, etc.).

Adaptive Layouts

The app must support two breakpoints:

- **Up to 840 dp** → mobile layout.
- **From 840 dp and above** → wide-screen layout.

Technical Requirements

Main Screen — Weekly Report Entry

The Main Screen is updated to provide access to the **Weekly Activity Report**. A textual label “**Report**” is displayed inside the **Step Counter Card**.

Layout

- The “Report” label is placed on the right side of the Step Counter Card.
- The label is top-aligned relative to the primary step count value, visually associating it with the main metric.
- The label is informational only and does not have its own interactive state.



Interaction Behavior

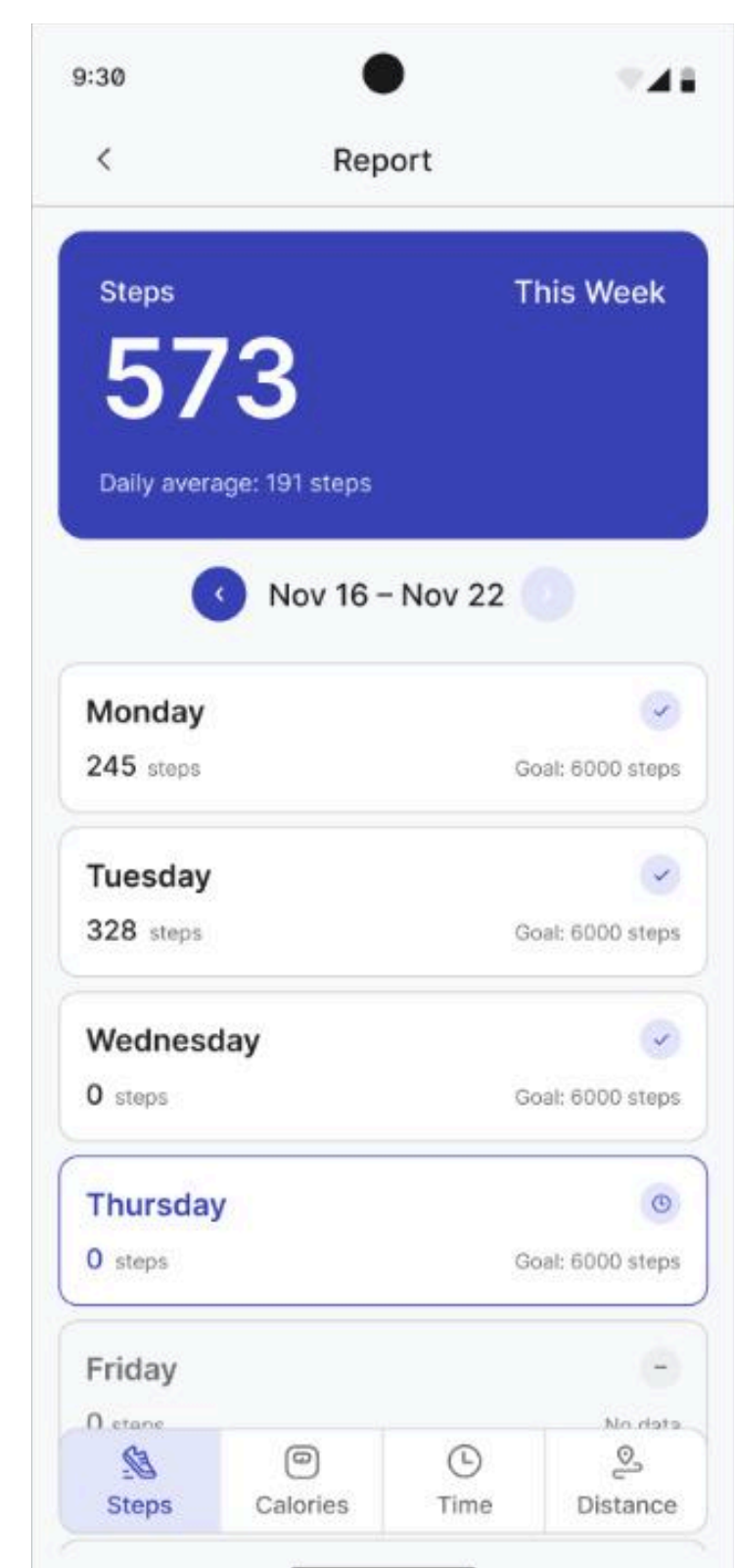
- The Step Counter Card acts as a general entry point to the Weekly Activity Report.
- Tapping on any **non-interactive** area of the card navigates the user to the Weekly Activity Report screen.
- Existing interactive elements inside the card **retain their original behavior**, including:
 - edit steps action,
 - play / pause step tracking.

Weekly Activity Report

The Report screen is designed to provide a detailed overview of the user’s weekly activity statistics. It allows users to analyze their progress for a selected week and compare daily values across different activity metrics.

The screen operates with four core activity metrics: Steps, Calories, Time, Distance. At any given moment, only one metric is active, determined by the user’s selection in the bottom tab bar.

All primary elements of the screen — including the summary card and the daily list — update dynamically based on the currently active metric.



Top Bar

The Top Bar is located at the top of the screen and contains:

- a Back button (arrow icon) on the left;
- the *Report* title centered.

Back button behavior:

- tapping the button returns the user to the Main Screen;
- the Report screen is removed from the back stack, without preserving its internal state.

Top Summary Card

Below the Top Bar, a primary summary card is displayed. This card presents weekly statistics for the currently active metric. The card provides a quick overview of overall progress for the selected week.

The Top Summary Card displays:

- the name of the active metric (**Steps / Calories / Minutes / Kilometers**);
- a primary numeric value representing the **total value for the selected week**;
- a secondary text label in the format: Daily average: X, where the unit depends on the active metric:
 - Steps — steps
 - Calories — kcal
 - Time — min
 - Distance — km

When the active metric is changed via the bottom tab bar:

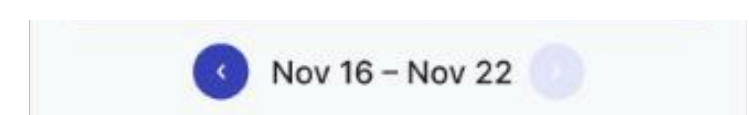
- the content of the Top Summary Card updates immediately;
- numeric formatting and units adjust according to the selected metric.

Week Selector

The Week Selector is located below the Top Summary Card and is used to navigate between weekly reports.

The selector consists of the following elements:

- a **previous week** button (left arrow);
- a date range label in the format: *Nov 16 – Nov 22*;
- a **next week** button (right arrow).



Week Selector Behavior

- By default, the **current week** is selected.
- Tapping the previous week button:
 - navigates to the previous calendar week;
 - updates the data shown in the Top Summary Card;
 - updates the list of days for the selected week.

- Tapping the **next week** button:
 - navigates to the next calendar week **only if it has already started**.
- If the next week **has not started yet**:
 - the next week button is **disabled**.

Date Range

- A week is defined as a **fixed calendar week**, starting on Monday and ending on Sunday.
- The displayed range always covers **7 consecutive calendar days** (e.g. Nov 16 – Nov 22).
- The week range is independent of the day when the user started using the app.
- If the user installs or starts using the app mid-week, the current calendar week is still shown, even if some days contain no data.

General Logic

- The selected week defines the **time range** for all data displayed on the screen.
- When the week changes:
 - the **Top Summary Card** updates to reflect aggregated data for the selected week;
 - the **daily list** updates to show values for each day of that week.
- The currently selected **metric tab** (Steps / Calories / Time / Distance) remains unchanged when switching weeks.
- Changing the week refreshes all displayed data **for the active metric only**, within the newly selected date range.

Weekly Breakdown List

This section displays a detailed breakdown of activity data for each day of the selected week.

Structure

- The list always contains **7 items**, one for each day of the week (Monday → Sunday).
- Days are ordered chronologically within the selected week.
- Each list item represents **one calendar day**.
- The **weekly list is scrollable**.

Day Item Content

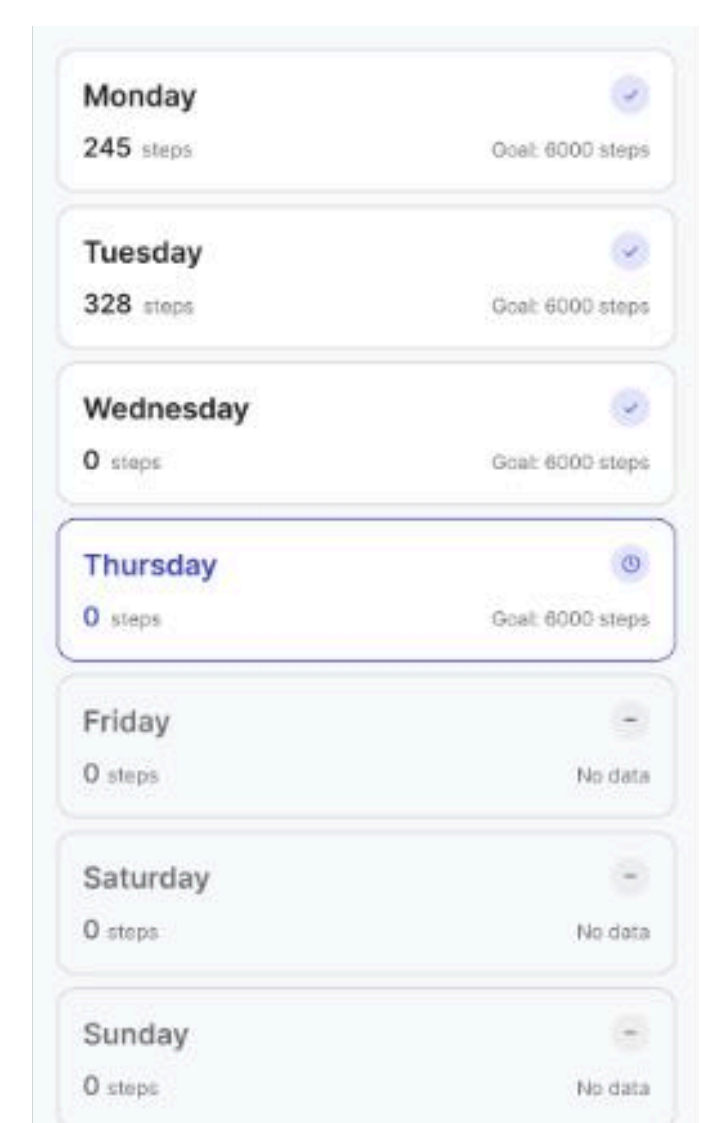
Each day item includes:

- Day name (e.g. Monday, Tuesday).
- Primary value based on the currently active metric
- Unit corresponding to the active metric.
- A status icon displayed in the top-right corner of the item.

Status Icons

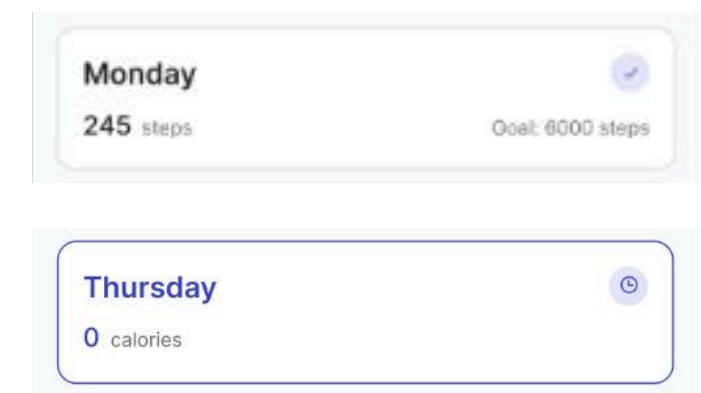
Each day item displays a status icon that reflects its state:

- ✓ Check icon – shown for completed past days that have recorded data.
- 🕒 Clock icon – shown for the current day of the current week and indicates that this day is still in progress.
- – Minus icon – shown for inactive days with no available data.



Goal Display

- The Goal label is shown only when the active metric is **Steps**.
Example: *Goal: 6000 steps*
- For all other metrics (Calories, Time, Distance), the Goal is **not displayed**.



No Data State

- If there is no data available for a given day, the item is displayed in an **inactive state**.
- This may occur when:
 - the day has not occurred yet (future days within the selected week), or
 - the app was installed mid-week, and no historical data exists for earlier days.
- In this case:
 - the primary value is shown as 0,
 - a “No data” label is displayed,
 - the item appears visually muted to indicate inactivity.
- The “No data” label is shown consistently for **all metrics**.

Active Day Highlight

- The **current day** (if it belongs to the selected week) is visually highlighted.
- This helps the user quickly identify today’s activity within the weekly context.

Interaction Behavior

- Day items are **not interactive**.
- The list is intended for **read-only overview** of weekly data.

Update Behavior

The weekly list updates automatically when:

- the active metric is changed via the bottom tab bar,
- the selected week is changed using the week selector.

Bottom Metric Tabs

The Bottom Metric Tabs component is a fixed navigation element located at the bottom of the Report screen. It is used to switch between different types of weekly analytics and defines the active metric for the entire screen.

Available Metrics

The Bottom Metric Tabs include four tabs: **Steps, Calories, Time, Distance**. Each tab consists of:

- a metric-specific icon;
- a text label representing the metric name.

Active Tab Behavior

- Only **one tab can be active at a time**.
- The active tab is visually highlighted (according to the design system).
- The active tab determines which metric data is displayed across the Report screen.

Impact on Report Screen Content

When the user selects a different tab:

- **Top Summary Card** updates immediately:
 - the metric title changes;
 - the primary value (weekly total) updates;
 - the Daily Average value and its unit update accordingly.
- **Weekly Breakdown List** updates:
 - daily values reflect the selected metric;
 - additional elements appear or disappear depending on the metric (for example, Goal is shown only for Steps).
- **Week Selector** remains unchanged and keeps the currently selected week.

All visible data on the screen always corresponds to the currently active metric.

Navigation Behavior

- Switching tabs:
 - does **not** open a new screen;
 - triggers an immediate in-place content update.
- The selected tab remains active until another tab is chosen.

Layout and Scrolling

- Bottom Metric Tabs are **fixed to the bottom of the screen**.
- They do not scroll with the content.
- Only the Weekly Breakdown List is **scrollable**.
- The Top Summary Card and Week Selector remain visible while scrolling the list.

Default State

- When the Report screen is opened for the first time, **Steps** is selected by default.
- All Report screen components are initialized using the **Steps** metric.

Report Screen — Wide Screen Adaptation

On wider devices (tablets, large phones, landscape mode), the Report screen preserves the **same logic, structure, and hierarchy** as the mobile version. The differences are limited to **layout composition and width constraints**, without introducing new behaviors.

General Principles

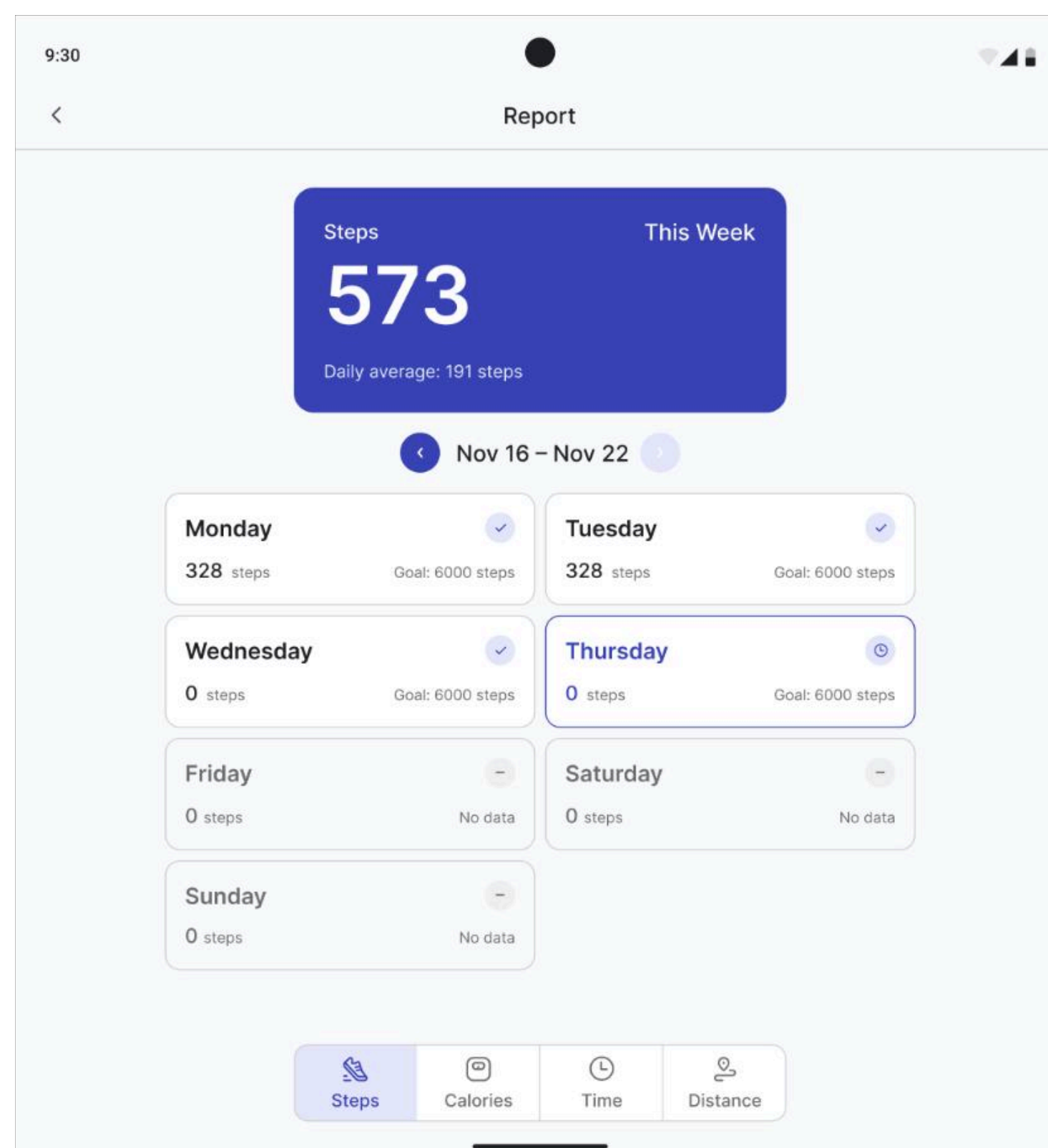
- All primary screen elements are **center-aligned**.
- The vertical order of elements remains unchanged.
- Navigation, interactions, and screen behavior are identical to the mobile layout.

Fixed Containers

- The Top Summary Card, Week Selector, and Bottom Metric Tabs are centered horizontally.
- Each of these components uses a **fixed width of 394 dp**.
- These elements do not stretch to fill the screen width, regardless of device size.

Weekly Day List

- The list of days is displayed in **two columns**.
- The entire list is placed inside a centered container with a maximum width of **600 dp**.
- Each Day Item keeps the same visual design, states, and behavior as defined for the mobile version.
- Vertical scrolling applies only to the **weekly day list**, while the Top Summary Card, Week Selector, and Bottom Metric Tabs remain static.



Useful Links for This Challenge

- [Gemini API quickstart](#)
- [Gemini API](#)
- [Monitor connectivity status](#)
- [Read network state](#)
- [Foreground services overview](#)
- [Foreground Services video](#)
- [About notifications](#)
- [Guide to app architecture](#)
- [Create a Splash Screen](#)
- [UX With Material3](#)
- [Full Guide to Material3 Theming](#)
- [Request runtime permissions](#)
- [Bottom sheets](#)
- [DataStore](#)
- [Physical Activity Permission](#)
- [Ignore Battery Optimization](#)
- [Save data in a local database using Room](#)
- [The Full Jetpack Compose Responsive UI Crash Course](#)
- [How to Save & Restore the Scroll Position of a LazyColumn Persistently](#)
- [Stateful vs. Stateless Composables](#)
- [State Hoisting in Compose](#)
- [Managing State in Jetpack Compose \(Codelab\)](#)